Shri Sant Gajanan Maharaj, Shegaon

Our Divine Source of Inspiration

SHRI SANT GAJANAN MAHARAJ

COLLEGE OF ENGINEERING, SHEGAON

Affiliated to Sant Gadge Baba Amravati University, Amravati

Technology That Removes the Middleman

DISTRIBUTED LEDGER TECHNOLOGY (8KS04)

Dr. N. M. Kandoi

Spring-Semester 2025-26

Course Objective

The learning objectives of this course are to:

- Develop a foundational understanding of distributed ledger technology, including its technical and institutional design elements.
- Analyse the efficiency, safety, and broader financial market implications of distributed arrangements using an analytical framework.
- Explore cryptographic principles and their application in cryptocurrencies, focusing on hash functions, digital signatures, and simple cryptocurrency structures.
- Examine the design, consensus mechanisms, and operational intricacies of Bitcoin, including decentralization and proof-of-work.
- Gain practical knowledge of Bitcoin mechanics, storage methods, mining processes, and associated ecological and economic implications

Course Outcomes

Upon completion of the course, students will have the ability to:

- Understand the foundational principles and technical design of distributed ledger technology and its operational frameworks.
- Evaluate the implications of distributed systems on efficiency, safety, and financial markets using analytical tools.
- Apply cryptographic concepts like hash functions and digital signatures to comprehend cryptocurrency structures.
- Demonstrate knowledge of Bitcoin's decentralized architecture, consensus mechanisms, and proof-of-work model.
- Analyze Bitcoin's transaction mechanics, storage solutions, and network infrastructure.
- Assess Bitcoin mining processes, including hardware, energy consumption, ecological impact, and incentive strategies

S.S.G.M. College Of Engineering Shegaon
Department of Computer Science & Engineering

Vision:

To achieve global recognition in Computer Science and Engineering education to serve the growing needs of industry and society.

Mission:

- To imbibe programming skills in latest areas of computer science through project work
- To prepare the Innovators, Entrepreneur, Researchers and Educationalist through quality education and Industry Institute Interaction
- To prepare good human beings who work with missionary zeal for upliftment of society.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

Program Educational Objectives (PEOs) of the department were established through consultation process amongst stakeholders and are set as follows

- To excel in communication skill, teamwork, multidisciplinary approach and an ability to relate engineering issues to broader social context.
- Graduate will demonstrate the skills of Innovation, Entrepreneurship and Research.
- Graduate will be Computer professionals with latest skills and technologies to cater the needs of global IT industry and National Organization.

PROGRAM SPECIFIC OUTCOMES (PSO's)

- Demonstrate knowledge of advanced programming techniques to solve contemporary issues.
- Perform projects with novel ideas and implement them.
- Understand and develop ethical work culture and attitude by participating in social service.

Books-Textbook and Reference Books

Book	Title	Author	Publication
Text-Book	Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction	Andrew Miller, Arvind Narayanan, Edward Felten, Joseph Bonneau, and Steven Goldfeder.	Princeton University.

Book	Title	Author	Publication
R-1	Mastering Blockchain: Distributed ledger technology, decentralization, and smart contracts explained	Imran Bashir	Packt Publication
R-2	**Distributed ledger technology in payment, clearing and settlement - An analytical framework**	**----**	**Bank for International Settlements -Pub**

Unit I: R-2 Ch-1 and 2, **Unit II: R-2 Ch-3**

Unit III: TextBook-Ch-1 **Unit IV: TextBook-Ch-2**

Unit V: TextBook-Ch-3 and 4 **Unit VI: TextBook-Ch-5**

Contents- Unit-I **Distributed Ledger Technology**

Distributed Ledger Technology:

- Introduction, Background
- Technical design elements**(How DLT Works - System Functionality**)
  - Maintaining information on the ledger
  - Updating the ledger
    - Validation and consensus
    - *Technical roles of nodes (differentiated or not differentiated)*
  - Process flow
- Institutional design elements(**How DLT Is Governed- control and governance.)**
  - Operation of the arrangement
  - Access to the arrangement (unrestricted or restricted)
- Potential configurations and trade-offs

Analytical framework:

- Understanding the arrangement
- Potential implications for efficiency
- Potential implications for safety
- Potential broader financial market implications

Framework: General structure to guide the implementation and operation of DLT

Analytical Framework: Detailed approach to systematically evaluate the technical, operational, and institutional aspects of using DLT in payment, clearing, and settlement systems.

Analytical framework :Specialized type of framework explicitly designed for analysis and evaluation

Framework : Broader and can be used for planning, organizing, or guiding any process or system.

Introduction to Cryptography & Cryptocurrencies:

- Cryptographic Hash Functions, SHA-256
- Hash Pointers and Data Structures
- Digital Signatures
- Public Keys as Identities
- A Simple Cryptocurrency

Contents- Unit-IV Bitcoin

Bitcoin:

- Centralization vs. Decentralization
- Distributed consensus
- Consensus without identity using a block chain
- Incentives and proof of work

Contents- Unit-V **Mechanics of Bitcoin**

Mechanics of Bitcoin:

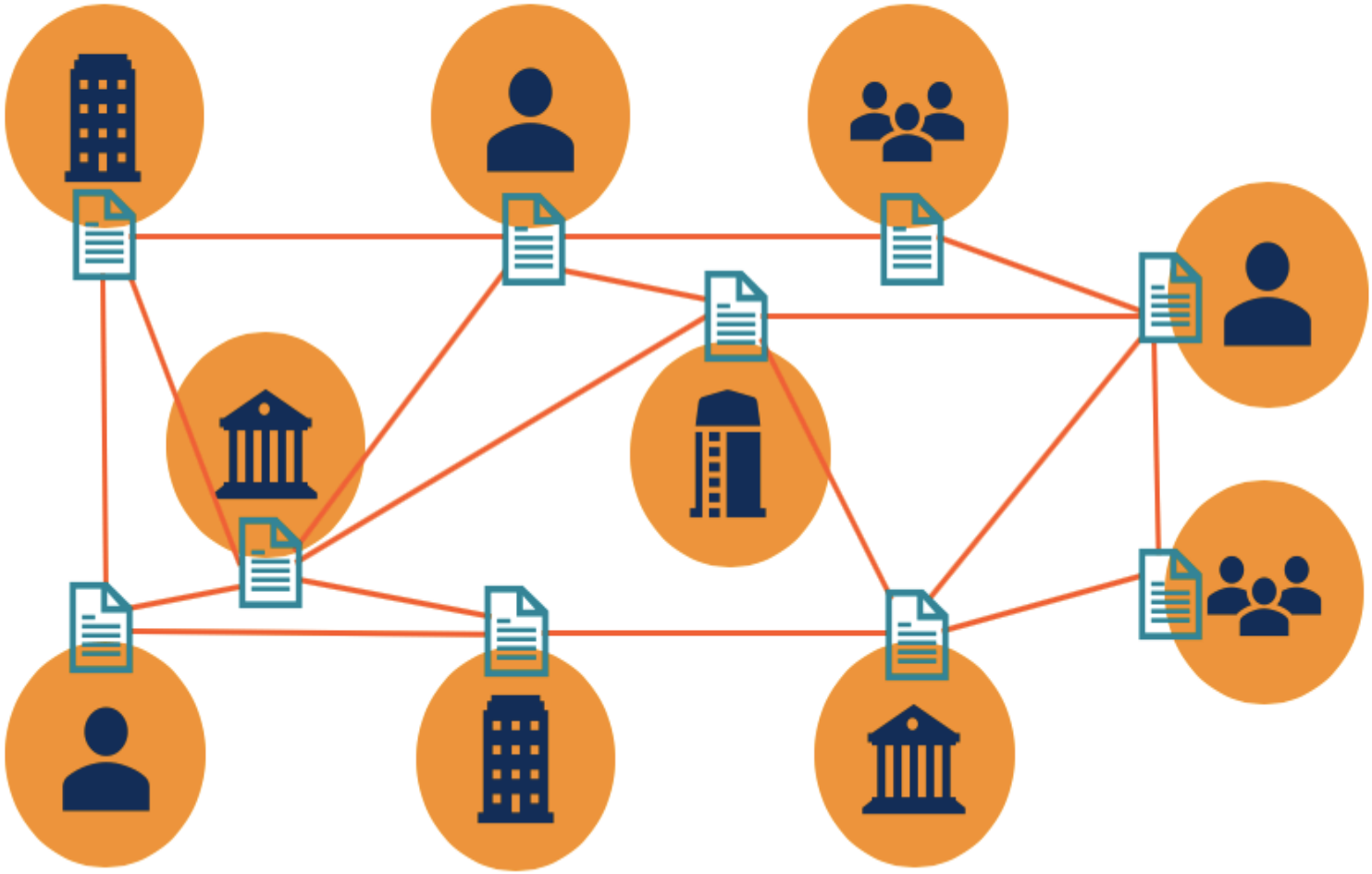
- Bitcoin transactions, Bitcoin Scripts, Applications of Bitcoin scripts
- Bitcoin blocks, Bitcoin network
- How to Store and Use Bitcoins
- Simple Local Storage, Hot and Cold Storage
- Splitting and Sharing Keys
- Online Wallets and Exchanges
- Payment Services, Transaction Fees, Currency Exchange Markets

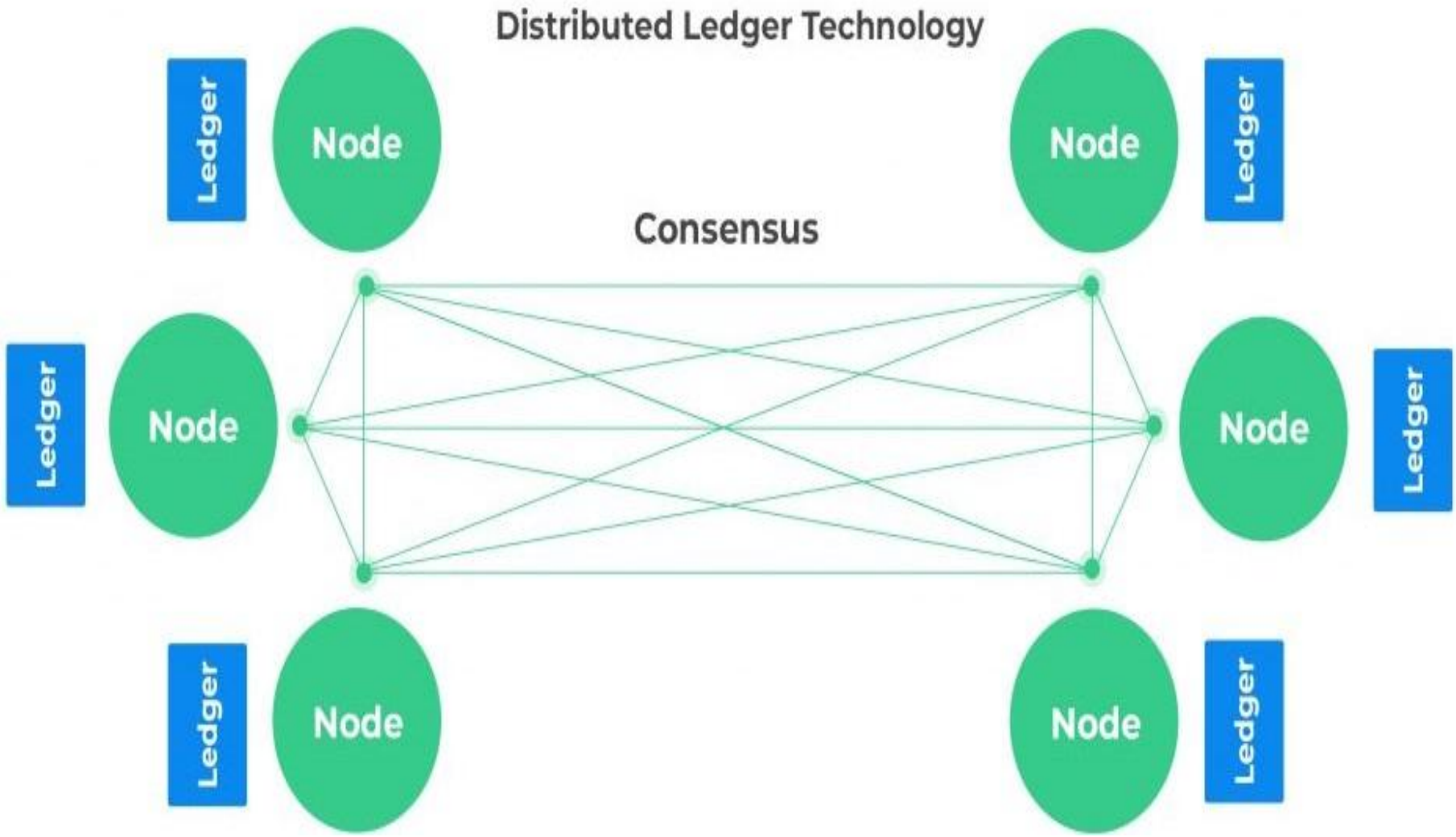
Contents- Unit-VI **Bitcoin Mining**

Bitcoin Mining:

- The task of Bitcoin miners
- Mining Hardware
- Energy consumption and ecology
- Mining pools
- Mining incentives and strategies

Distributed Ledger Technology




Distributed Ledger Technology
Ledger
Node
Node
Ledger
Consensus
Ledger
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Ledger

The Properties of Distributed Ledger Technology (DLT)

Programmable
A blockchain is programmable (i.e. Smart Contracts)

Distributed
All network participants have a copy of the ledger for complete transparency

Secure
All records are individually encrypted

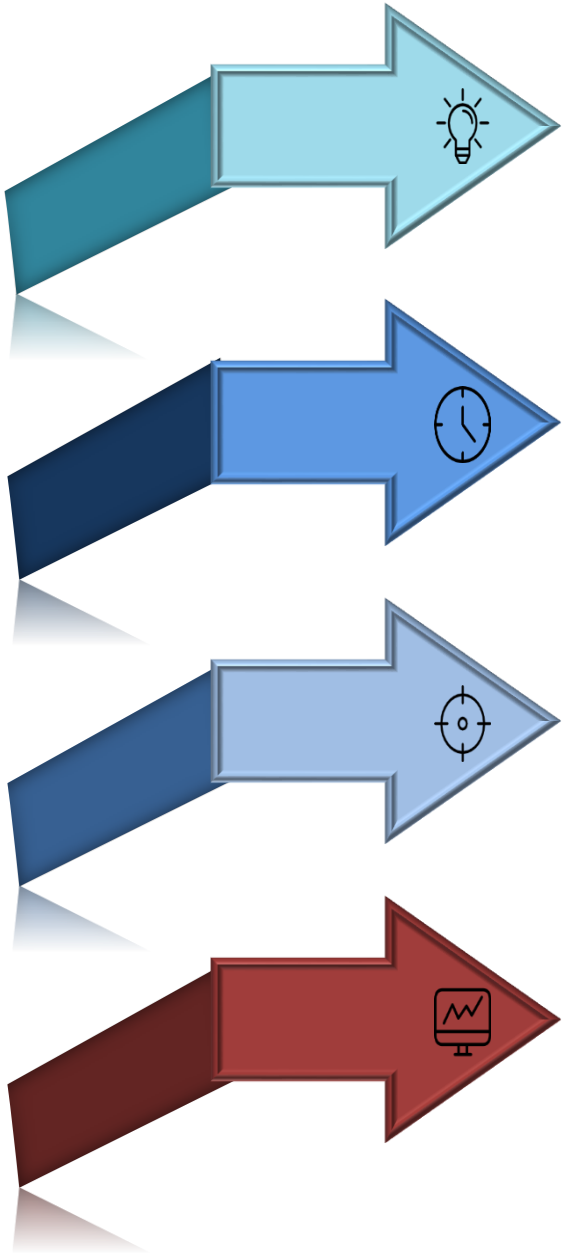
Immutable
Any validated records are irreversible and cannot be changed

Anonymous
The identity of participants is either anonymous or pseudonymous

Unanimous
All network participants agree to the validity of each of the records

Time-stamped
A transaction timestamp is recorded on a block

Key Learning Outcomes of the Course



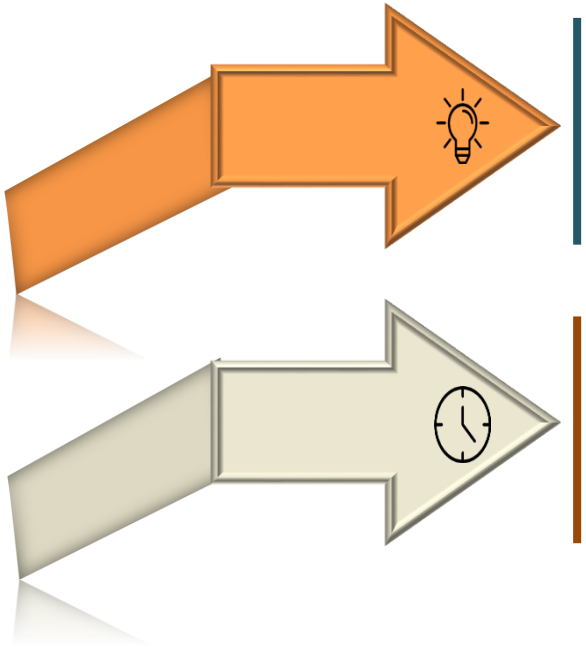
01 Understand the fundamentals DLT and blockchain systems

02 Analyze efficiency, safety, and financial market implications of DLT

03 Learn how cryptography secures digital transactions and cryptocurrencies

04 Gain knowledge of Bitcoin architecture, consensus mechanisms, and incentives

Key Learning Outcomes of the Course



05 Understand Bitcoin transactions, wallets, storage methods, and usage

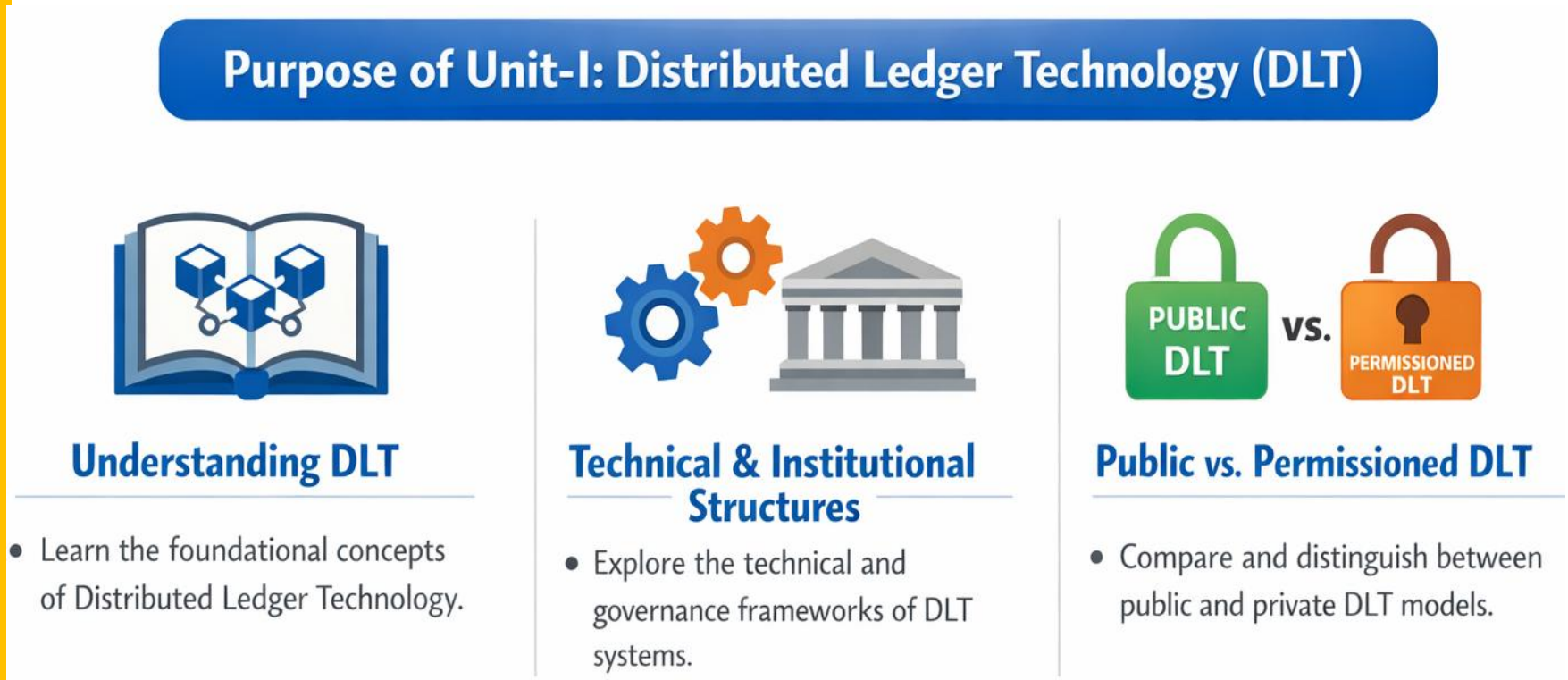
06 Learn the concepts, process, and impact of Bitcoin mining

Purpose of Unit–I:

To provide a foundational understanding of Distributed Ledger Technology.

To explain the technical and institutional structures of DLT systems.

To enable learners to distinguish between public and permissioned DLT models and understand their applications.



Introduction

- Concept of Distributed Ledger Technology and its role in modern digital systems.
- Impact of DLT on payment, clearing, and settlement processes.
- Features such as decentralization, transparency, and trust without intermediaries.

Background

- Evolution of distributed systems and databases.
- Need for DLT in financial and non-financial applications.
- Key developments leading to the adoption of DLT.

Technical Design Elements

- Core components: consensus mechanisms, cryptography, and data structures.
- Importance of security, scalability, and interoperability.
- Fault tolerance and data integrity in distributed environments.

Institutional Design Elements

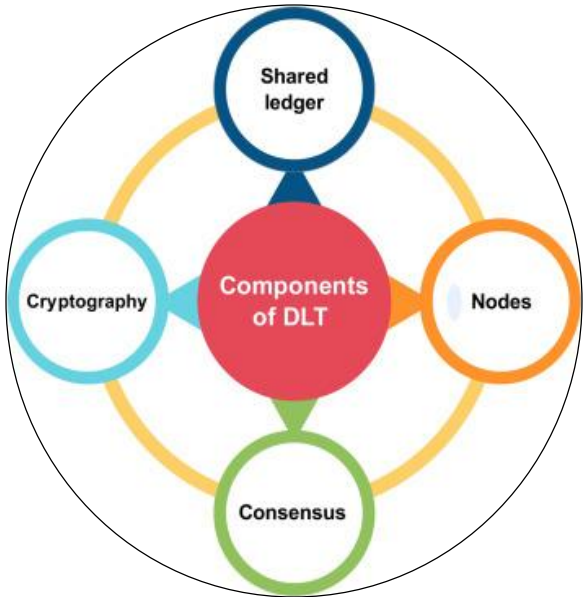
- Governance and operational frameworks of DLT systems.
- Roles of participants, validators, and administrators.

Operation of the Arrangement

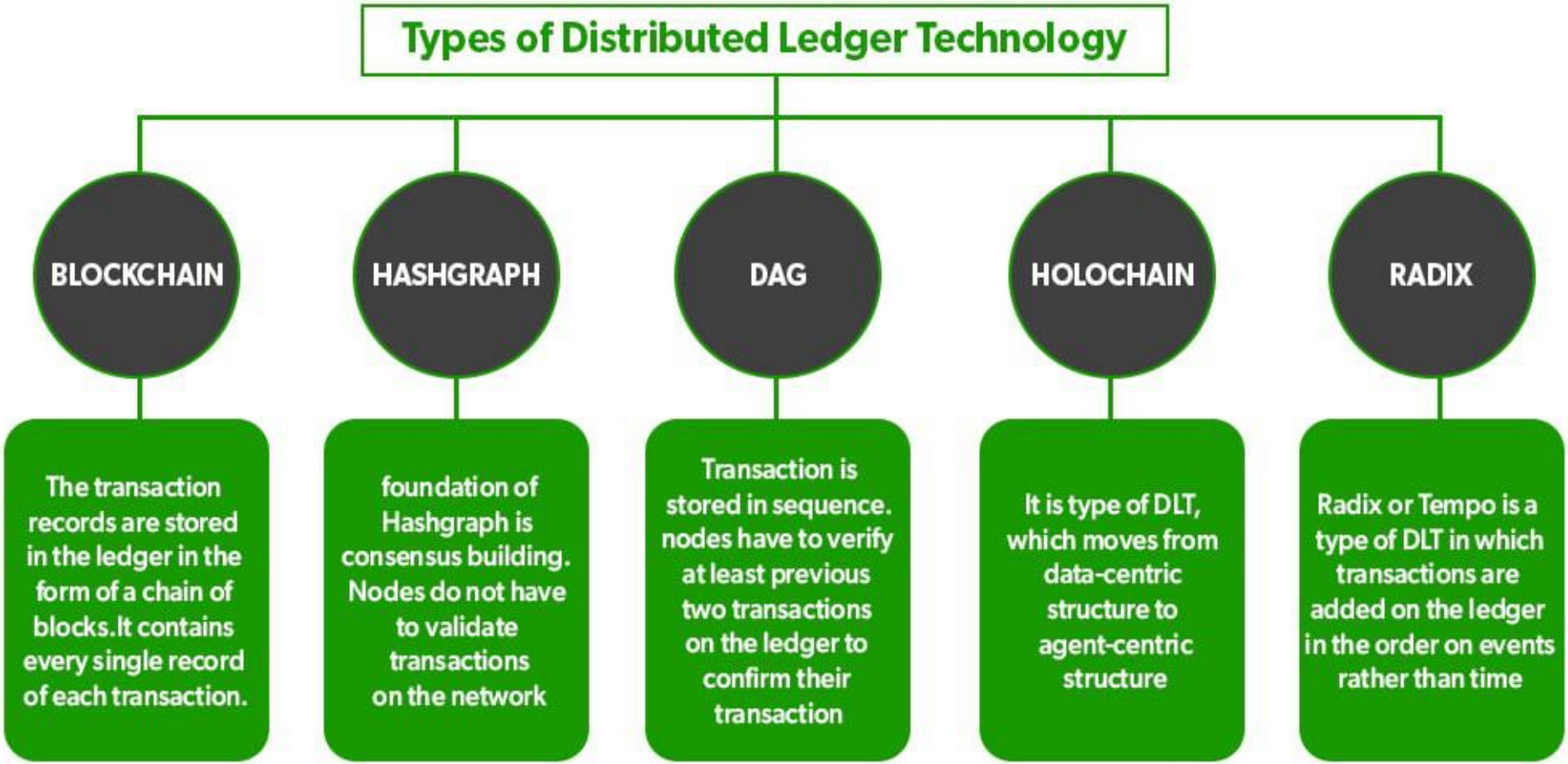
- Transaction creation, validation, and recording on the ledger.
- Mechanisms ensuring transparency, integrity, and efficiency.

Access to the Arrangement

- Unrestricted (Public DLT): Open participation without prior permission.
- Restricted (Private/Permissioned DLT): Access limited to authorized participants.



Different DLT architectures and how they record and validate transactions differently.



Distributed Ledger Technology

Distributed Ledger Technology (DLT)

What is DLT?

System where **the same record (ledger) is shared and updated across many computers (nodes)** instead of being stored in one central place.

It is also known as a “shared ledger” or simply distributed ledger.

No single person or authority controls it — **everyone has a copy, and all copies stay in sync**.

Real-Life Example :- Imagine a class expense notebook:

Your class decides to maintain a **shared notebook** for class expenses.

Every student has a **photocopy of the same notebook**.

When someone spends money, it is **announced to the whole class**.

Everyone writes the same entry in their notebook.

If someone tries to change an old entry, **others will notice immediately**.

➡ This shared notebook system is **Distributed Ledger Technology**

Distributed Ledger Technology?

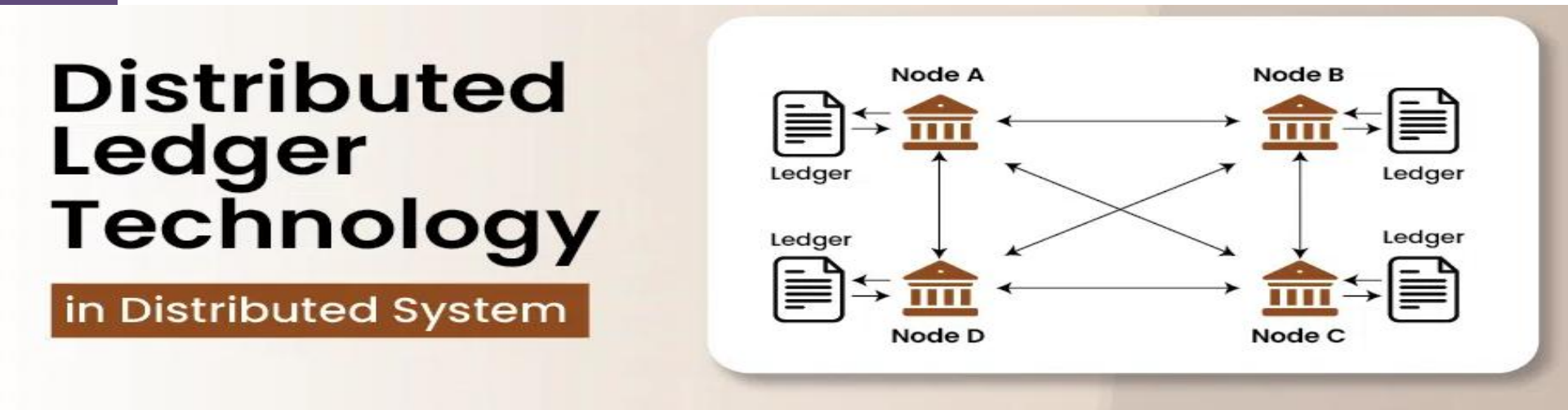
Distributed ledger :

- Unlike traditional databases, DLT does not rely on a single central location to store data.
- This absence of central control is the key factor that differentiates DLT from conventional databases.
- Decentralization enhances **security, transparency, and trust** among participating parties.
- A distributed ledger is a shared record of transactions that is **secured using cryptography** and **validated by network nodes through consensus**.
- DLT systems can be **decentralized or partially centralized**, depending on their design.
- **Blockchain** is one of the technologies under the umbrella of DLT.
- While blockchain is a method to implement a distributed ledger, **not all distributed ledgers are based on blockchain**.

DLT -- Distributed Ledger Technology, also known as a "shared ledger" or distributed ledger.

Distributed Ledger Technology?

- DLT enables users to **record, share, and synchronize data and transactions** across a distributed network with multiple participants.
- It represents a **family of technologies** with similar underlying structures but implemented in different ways and governed by different rules (blockchain is one such example).
- DLT can be **public or private**, depending on whether the ledger is accessible to everyone or only to selected participants.
- It can also be **permissioned or permissionless**, based on whether participants need authorization from a specific entity to access or update the ledger.
- By eliminating the need for a central authority, DLT **improves transaction speed** and **reduces transaction costs**.



Traditional vs Modern Distributed Ledger Systems

Traditional Distributed Ledger:

- In a traditional distributed database, a **system administrator** is responsible for maintaining consistency among multiple copies of the ledger.
- The simplest approach is to maintain a **master copy** of the ledger.
- This master copy is **periodically updated and manually shared** with all network participants.

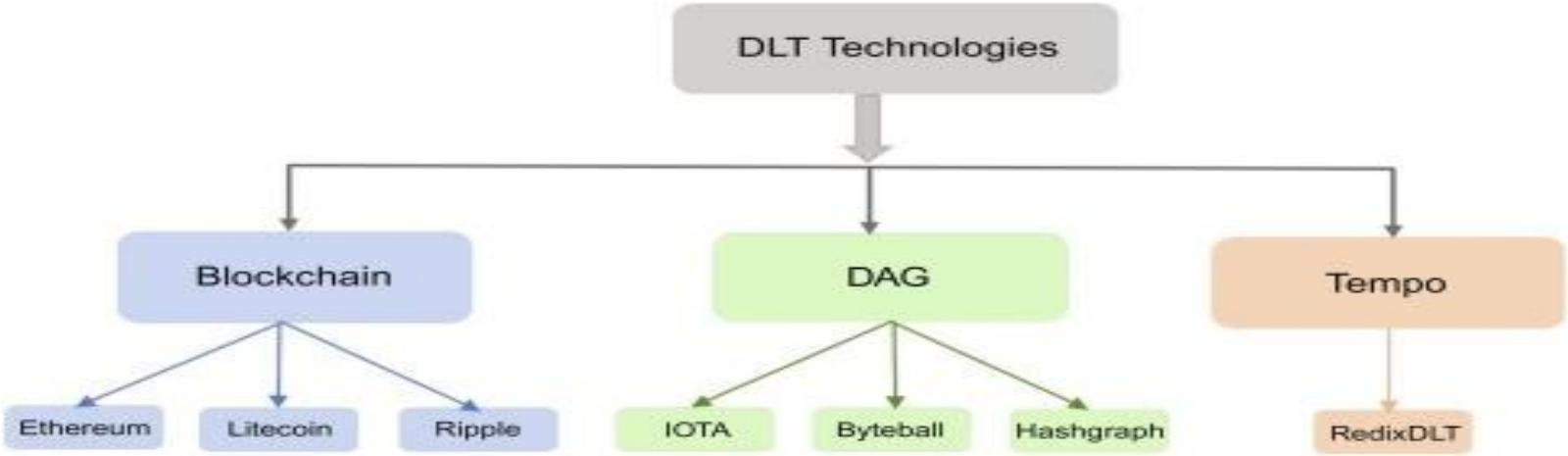
Modern Distributed Ledger (DLT):

- Modern DLT systems (e.g., **Bitcoin and Ethereum**) are designed to operate **without a trusted central authority**.
- Bitcoin maintains its distributed database in a **decentralized manner** using **cryptography and consensus-based validation**.
- Transactions are carried out in a **peer-to-peer** way and broadcast to all participants in the network.
- Participants validate transactions in groups called **blocks**.
- Since transactions are stored in **linked blocks**, this form of DLT is known as **blockchain technology**.

Distributed Ledger Technology?

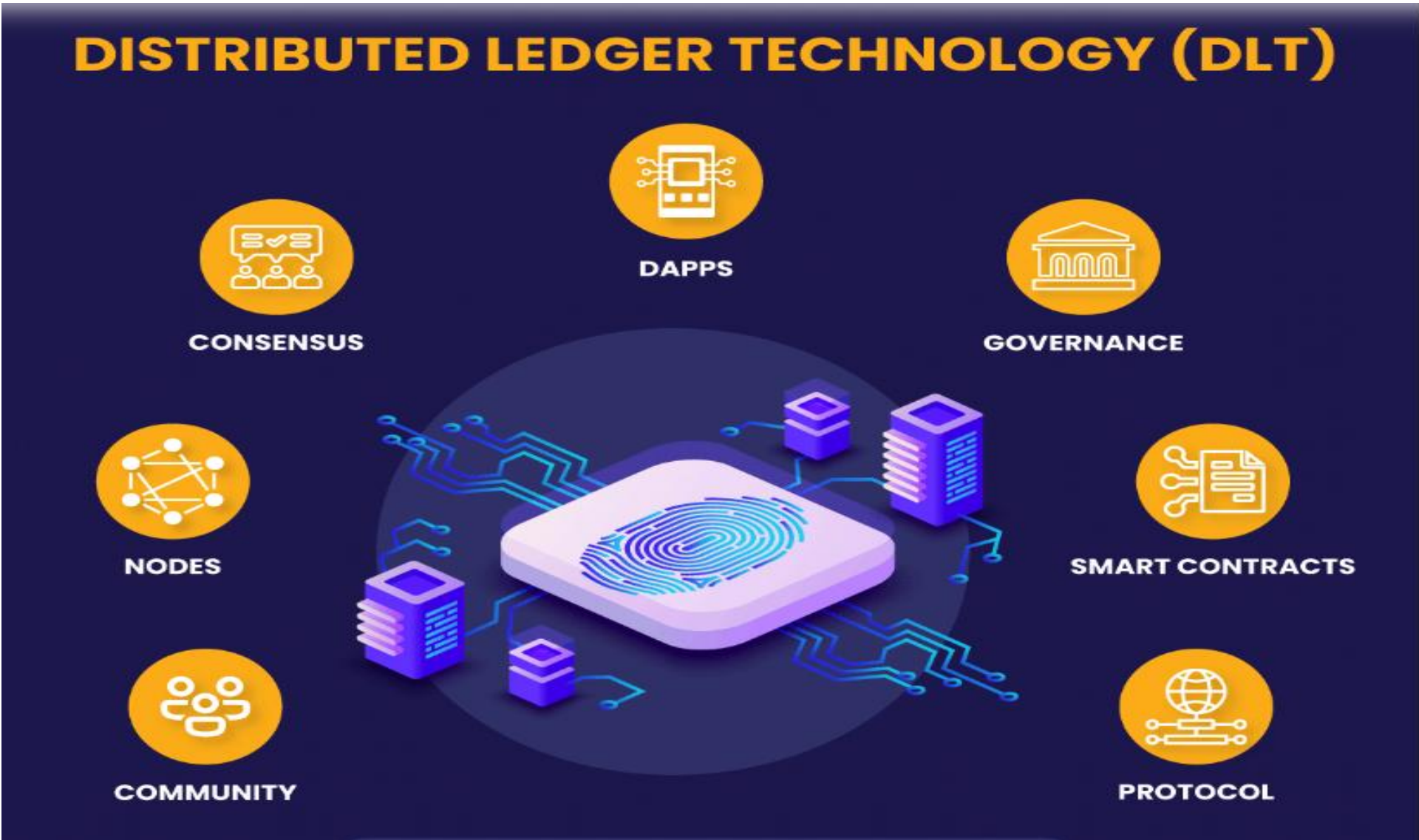
- DLT is a framework that supports several related technologies, including **blockchain**.
- In a DLT network, **each node maintains its own copy of the ledger**.
- Nodes can **directly transact with each other** without involving a central authority.
- There is **no need for mediation or permission** from a centralized system to access or update the ledger.
- After every transaction, the ledger is **updated and synchronized across all nodes**.
- Popular DLT platforms include **Bitcoin, Ethereum, Hyperledger Fabric, R3 Corda, and Quorum**.

With advances in cryptography and its application to ledger technologies, DLT is now well-positioned for mass adoption by both consumers and businesses.



Conceptual overview of Distributed Ledger Technology (DLT)

DLT works through the **combined role of nodes, consensus, smart contracts, governance, and community** to create a **secure, decentralized, and transparent system**.



DLT stands for distributed ledger technology. DLT is also known as the shared ledger or distributed ledger. It is a digital system that lets users and systems record transactions related to assets. It originated from peer-to-peer(P2P) network.

KEY FEATURES OF DLT

IMMUTABLE
Data once written, cannot be altered

DISTRIBUTED
Data stored in a distributed format which nodes having full or partial copies

SHARED
The ledger shared among peers, and central authority doesn't own it

APPEND ONLY
Data added to the ledger in append-only mode

TYPES OF DISTRIBUTED LEDGER TECHNOLOGIES

PERMISSIONED
Users need permission to access network features and functionality

PERMISSIONLESS
Users can join and use the network features and functionality without any permission

HYBRID
Hybrid is a combination of permissioned and permissionless

DLT ECOSYSTEM

HARDWARE
It is the hardware that participates in DLT functionality

BUSINESS
The stakeholders that make DLT possible

SOFTWARE
Software for different applications in a distributed ledger network

PROTOCOL
Protocol for the proper functioning of the DLT

Image provides an overview of DLT , its key features, types, and ecosystem components

Key Features of DLT:

1. **Immutable :** Once data is written, **it cannot be changed or deleted**.
2. **Distributed :** Data is stored across **many nodes**, each having full or partial copies.
3. **Shared:** The ledger is shared among participants; **no single owner** controls it.
4. **Append-Only:** New data is **only added**, not modified

Types of Distributed Ledger Technologies

- **Permissioned DLT:** Users need **approval** to access the network and mainly Used in banks, enterprises, and institutions.
- **Permissionless DLT:** Anyone can **join and participate freely** and example: Public blockchains like Bitcoin.
- **Hybrid DLT:** Combination of **permissioned and permissionless** and Some data is public, some is restricted.

DLT Ecosystem

- **Hardware:** Physical machines (nodes, servers) that run the network.
- **Software:** Applications and platforms built on DLT.
- **Protocol:** Rules that define **how the network works**.
- **Business:** Organizations and stakeholders using DLT for real-world use cases.

Introduction : DLT- Risk Factor ?

Understanding the Challenges of Distributed Ledger Technology

While DLT offers exciting possibilities, it also comes with certain challenges and risks.

Many risks related to payments, clearing, and settlement still exist—whether using a traditional centralized system or a synchronized distributed ledger.

Additionally, DLT can introduce new or different kinds of risks, such as:

- **Operational & Security Risks** – Potential weaknesses in the technology itself could be exploited.
- **Interoperability Problems** – DLT systems may not work smoothly with existing technologies and infrastructures.
- **Unclear Settlement Finality** – It can be difficult to confirm when a transaction is truly complete and irreversible.
- **Legal Uncertainties** – Questions remain about whether current laws fully support and protect DLT use.
- **Governance Gaps** – Many DLT systems lack strong, agreed-upon rules for decision-making and oversight.
- **Data & Privacy Issues** – Concerns around keeping data secure, private, and tamper-proof still need to be addressed

Technical Design Elements

Technical mechanisms that define how a DLT system stores data, verifies transactions, and updates the ledger securely across multiple nodes.

Use of Distributed Ledger Technology (DLT):

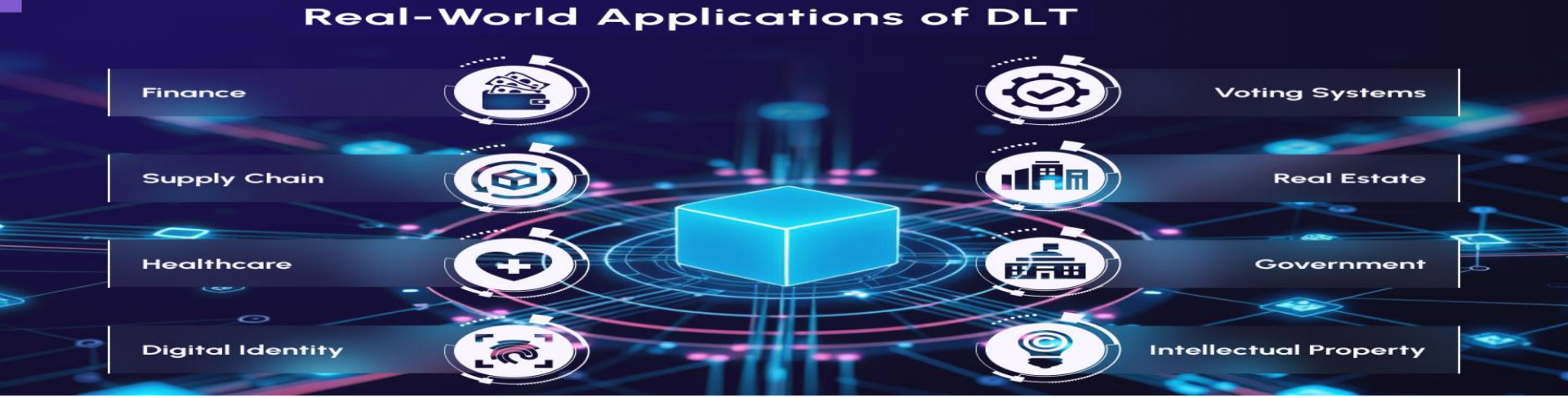
DLT is widely used for payment, clearing, and settlement activities. However, its implementation faces key challenges related to building systems that are **safe, secure, and scalable.**

Technical Design Elements of DLT:

DLT arrangements designed in multiple ways and may support some or all stages of a transaction flow. These arrangements primarily focus on two aspects:

How information is maintained on the ledger

How the ledger is updated



Core Function of Ledgers in DLT:

- Foundation for recording and managing information in decentralized systems.
- Used in applications like payments, clearing, and settlement to track ownership or balances of assets

Maintaining Information on the Ledger (DLT):

- The **ledger is the core component** of any DLT system and stores records and related information.
- In **payment, clearing, and settlement**, the ledger records **ownership or balances of digital assets**.
- **Native assets (native tokens):** Digital assets created directly on the ledger.
- **Non-native assets (non-native tokens):** Digital representations of physical or off-ledger assets.
- DLT systems use a **digital ledger** that keeps a summary of **transactions or account balances** of participant

Ways of Record Keeping:

- Some ledgers store a **complete history of all transactions**.
  - Example: **Blockchain**, where transactions are grouped into **blocks** and linked to previous blocks.
- Other ledgers store only **current account balances**, similar to **traditional bookkeeping**.
- Not all distributed ledgers use **blocks or chains**; blockchain is just one type of DLT.

Additional Information Stored on Ledgers:

- Ledgers may also store **financial contracts** or copies of contract terms.
- Some DLT systems support **smart contracts**, which are **self-executing programs** stored on the ledger.
- Smart contracts can automatically:
  - Make payments on specific dates
  - Distribute or collect funds based on events
  - End contracts automatically when conditions are met

This makes DLT efficient, transparent, and suitable for automation

Different DLT different way of maintaining information on Ledger

- Transactions grouped into blocks, and each block is linked to previous one using cryptographic hashes.
- This ensures immutability and security, as altering any block would break the chain.
- In Hyperledger(Private blockchain-Type of DLT) data is stored **in a private database on peer nodes of authorized organizations** and is accessed from chaincode on these authorized peers
- DLT **allows for storage of all information in a secure and accurate manner using cryptography**.
- Later same can be accessed using "keys" and cryptographic signatures
- Once the data is saved, it becomes an immutable database governed by the network's rules

Technical Design Elements : Updating the ledger

Here we will explore two points

- Validation and Consensus
- *Technical roles of nodes (differentiated or not differentiated)*

Distributed Ledger Updates

- In DLT Ledger updating is done by a group of authorized or selected nodes, and the updated ledger is then synchronized across all nodes.
- These nodes may be located across different sites, institutions, or even countries.
- Each node maintains an identical copy of the ledger.
- All nodes are connected and participate in updating and validating ledger changes.
- Based on the system's rules, any update made to the ledger is reflected across all copies within a specific time period, known as latency.

DLT: Technical Design Elements

Updating the ledger : Validation and consensus

Maintaining updated copy of digital ledger , across the multiple nodes in p2p n/w , is very challenging.

This can be possible using validation process and various types of consensus protocol.

Validation and Consensus

Cryptography : Cryptographic tools, public key cryptography and public key infrastructure, play an important role in DLT by identifying and authenticating

Approved participants, confirming data records and facilitating consensus on ledger updates.

Participants proposing changes to the ledger, authenticate themselves by providing their cryptographic digital signatures for the proposed change.

Validators will use cryptographic tools to

- Verify whether the participant has the proper credentials to do so.
- Restrict access to data so that only approved parties can see the information..

DLT: Technical Design Elements

Validation and Consensus – continue …

Consensus.

It is process by which nodes in a n/w agree on common state of the ledger.

This process relies on cryptographic tools, a set of rules or procedures reflected in the protocol

Consensus generally involves two steps

- **Validation**: Each validator identifies state changes that are consistent according to the rules of arrangement . For this each validator needs to rely on a record of previous states, either as a "last agreed state" or as a "chain of previous states".
- Agreement on ledger updates: Nodes agree to state changes to the ledger.

This stage of consensus process involves mechanisms or algorithms that resolve conflicting changes to the ledger.

DLT: Technical Design Elements

Technical design elements

- Maintaining information on the ledger
- Updating the ledger
  - Validation and consensus
  - ***Technical roles of nodes (differentiated or not differentiated)***

DLT – Collection of multiple nodes , and these nodes may play a variety of technical roles.

Examples : Various nodes and their roles---

• **System administrator:** Gatekeeper that controls access to system and provides certain services for the arrangement, including notary function, dispute resolution, standard-setting and regulatory reporting.

• **Asset issuer**: Issues new assets to participants in the system.

• **Proposer:** Proposes updates to the ledger.

• **Validato**r: Validates proposed updates to the ledge.

• **Auditor**: View the ledger but not make updates

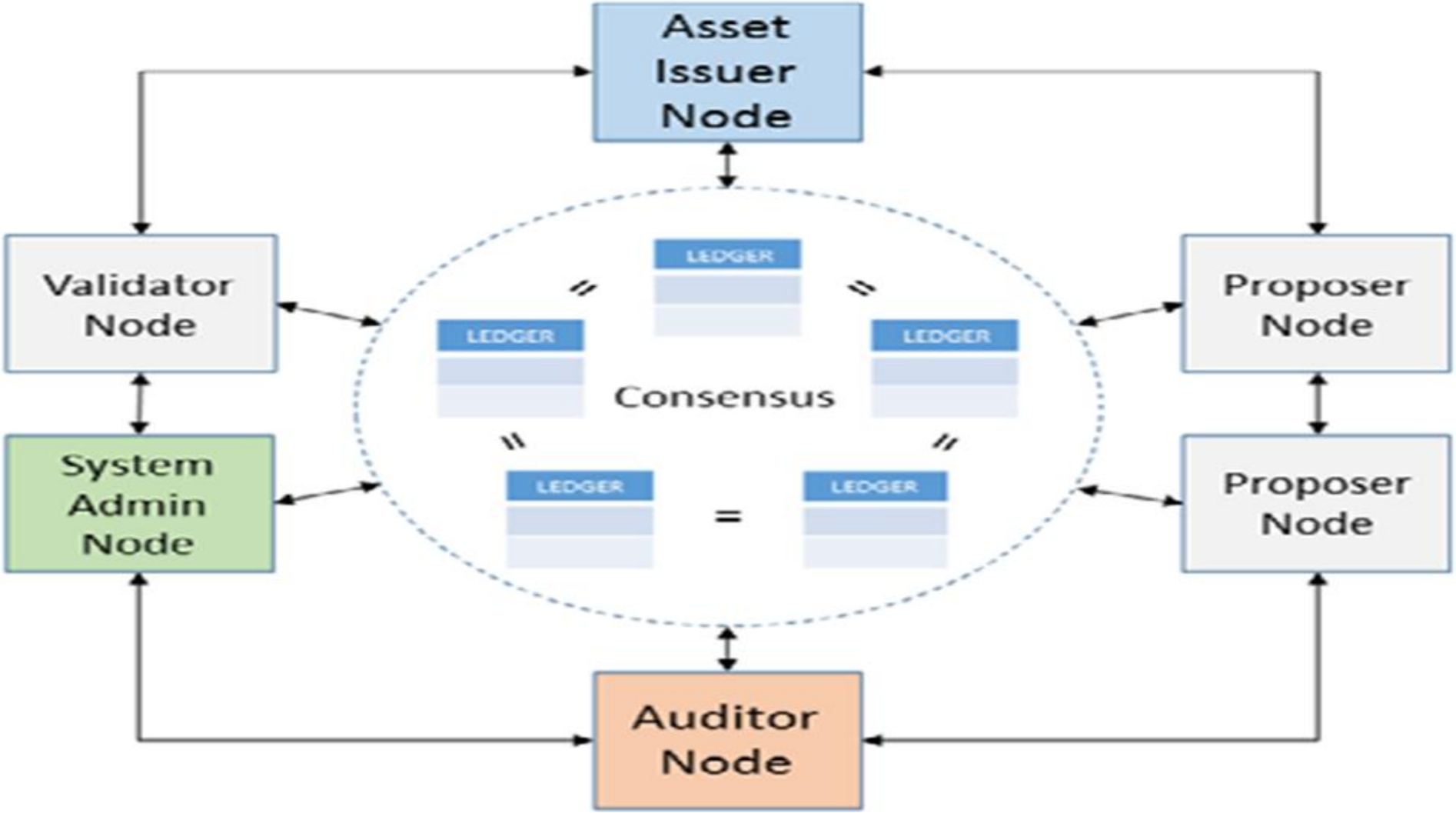


Image showing how a **DLT** system works using different types of nodes.

Ledger & Consensus (Center):

All participating nodes keep **identical copies of the ledger**. Through **consensus**, they agree on updates so every ledger remains the same.

Proposer Nodes: Create and submit new transactions to the network.

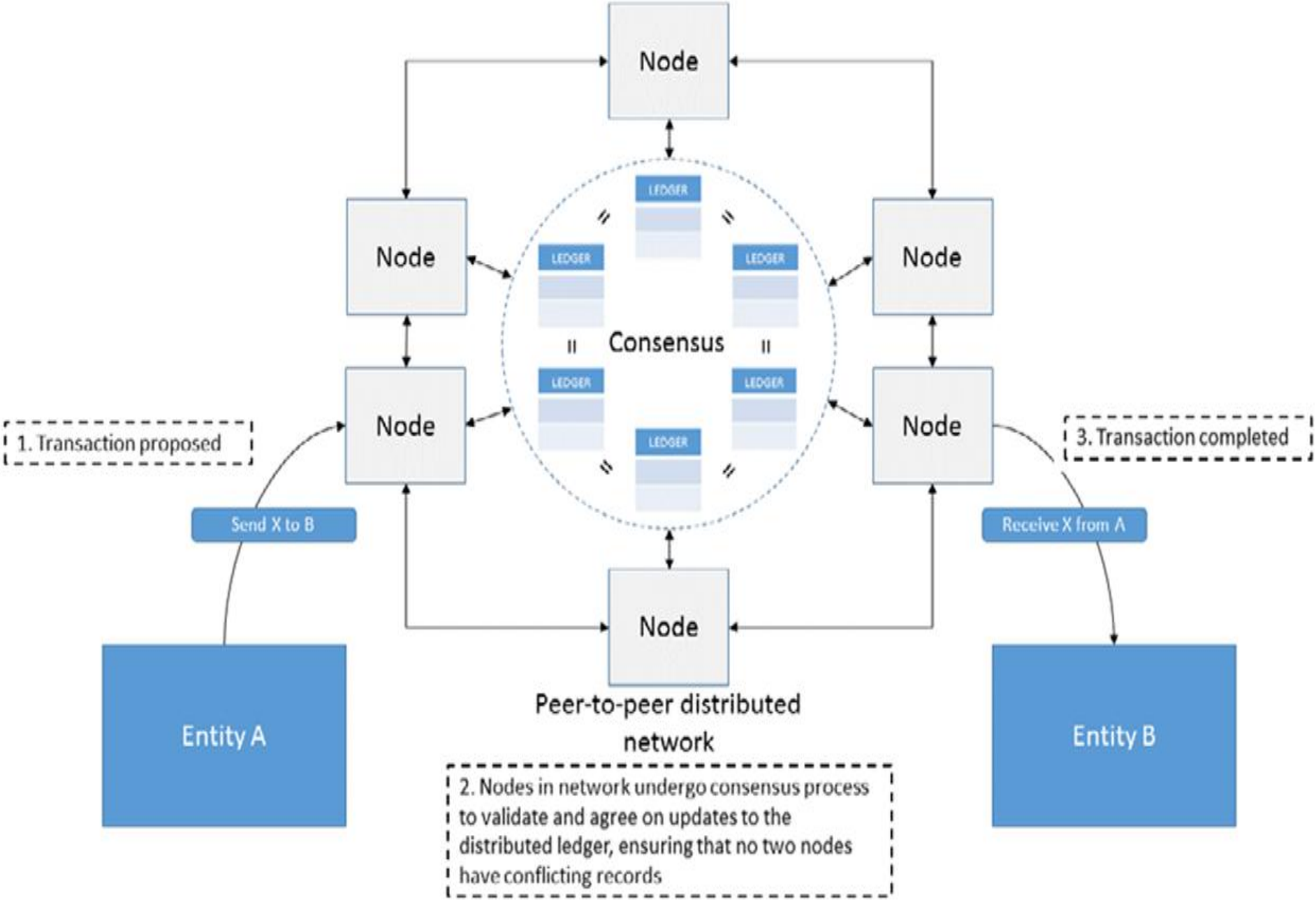
Validator Node: Checks whether proposed transactions are valid before they are added to the ledger.

Asset Issuer Node: Issues or creates digital assets (e.g., tokens, certificates) on the ledger.

System Admin Node: Manages permissions, network rules, and node access.

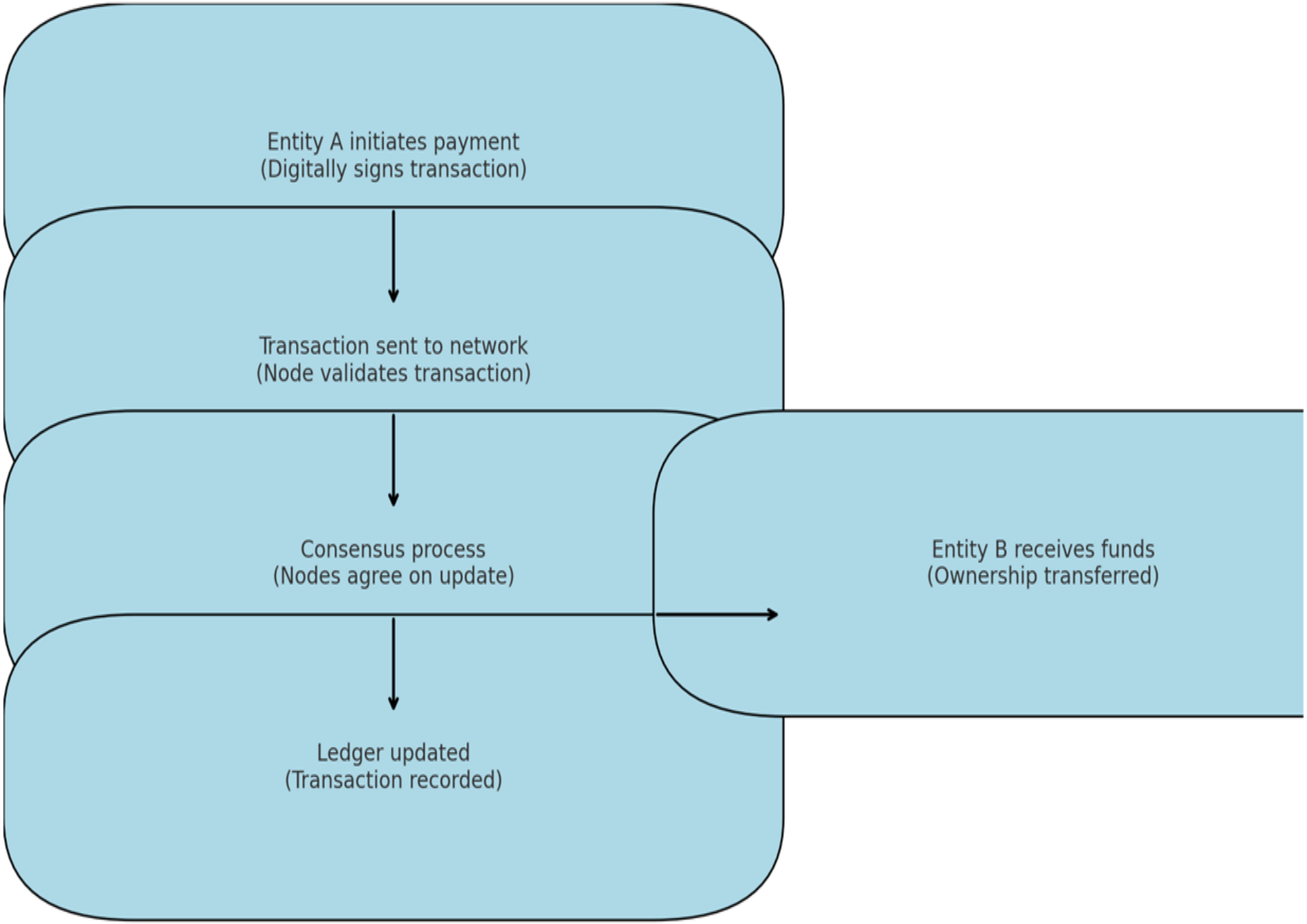
Auditor Node: Monitors transactions and ledger records for compliance and transparency.

Different nodes have different roles, but all work together through consensus to maintain a secure, shared, and synchronized ledger without a central authority.



Process Flow of a DLT-based payment system

Process Flow of a DLT-Based Payment System



DLT: Technical Design Elements - Process Flow

In this example, transaction process involves three major steps:

Transaction Process in DLT (Short)

1. **Initiation**

   Entity A initiates a payment by digitally signing a transaction to transfer funds to Entity B using cryptographic tools.

2. **Validation and Consensus**

   Network nodes verify Entity A's identity, check authorization and sufficient balance, and collectively agree on the transaction through a consensus process.

3. **Ledger Update**

   After approval, the ledger is updated and control of the asset is transferred to Entity B, who can initiate future transactions using their cryptographic credentials.

DLT transactions are digitally signed, validated through consensus, and finalized by updating the distributed ledger.

DLT: Institutional Design Elements in DLT

Institutional Design & Roles

Defines responsibilities for managing, accessing, and operating the DLT system.

Operation of the Arrangement

A single entity or multiple entities can manage the system.

Responsibilities include updating protocols, granting access, and maintaining the ledger.

Access to the Arrangement

Unrestricted: Open to all, with no access restrictions. Nodes may not know each other.

Restricted: Controlled access, with governance rules managed off-ledger.

Example of a Restricted DLT Arrangement

Entity A: System administrator and validator.

Entities B & C: Proposers who suggest changes but need approval from Entity A.

DLT: Institutional Design Elements

Institutional design – Continue …

Access to the arrangement (unrestricted or restricted)

Arrangements can be designed to accommodate any number of participants.

Arrangements may be unrestricted or restricted

Unrestricted arrangements :

No restrictions on access to the arrangement or to the roles of nodes within arrangement

Access is completely open, entities operating the nodes are not likely to know each other.

Difficult to govern such arrangement and entire set of rules governing interactions among nodes needs to be primarily “on-ledger” (encoded in the computer protocol).

Restricted arrangements :

Allow control over participants’ access to the arrangement.

Because access is controlled, the set of rules governing interactions can also be “off-ledger”.

When should a restricted DLT arrangement and when should an unrestricted arrangement be used?

No single best arrangement. Both **restricted** and **unrestricted** DLT arrangements are useful depending on the situation and requirements.

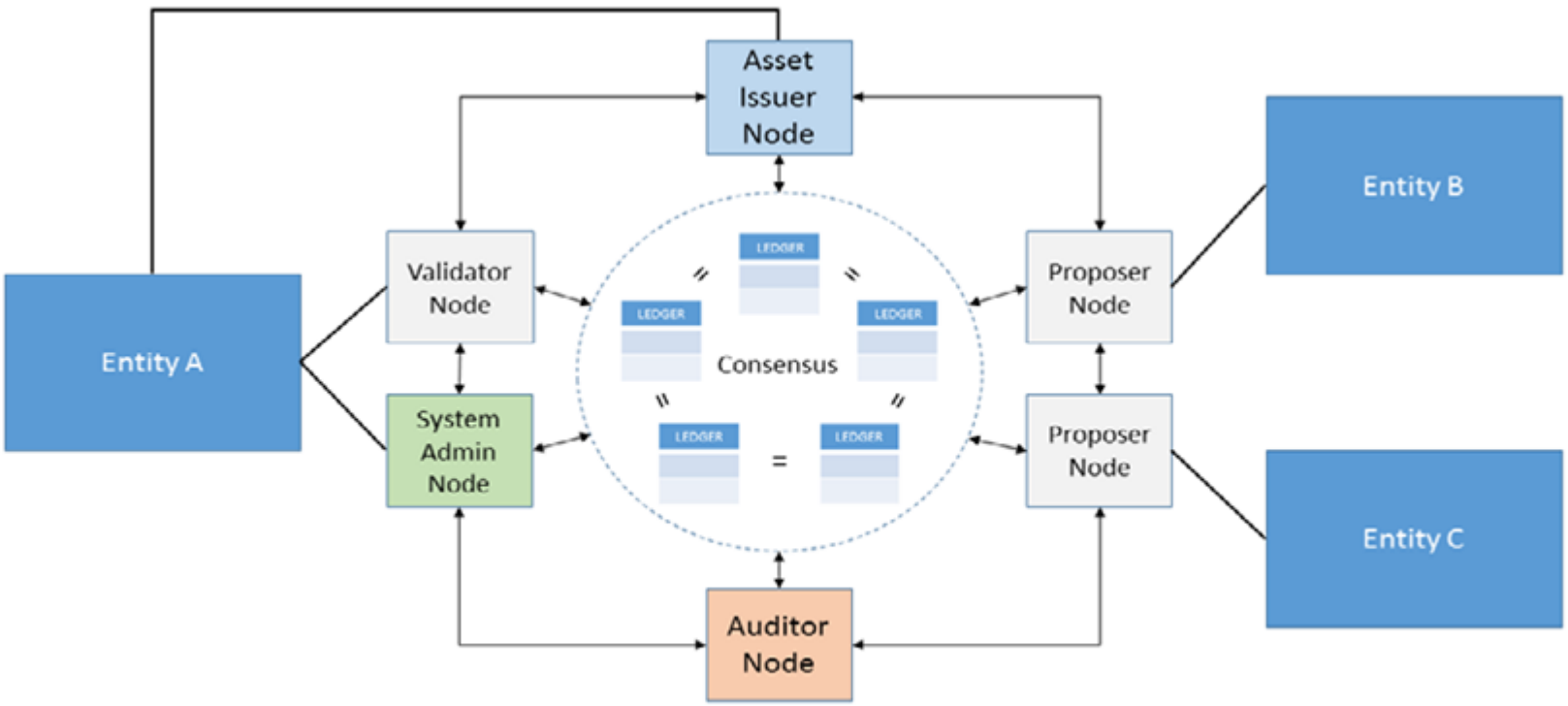
Restricted Arrangement	**Unrestricted Arrangement**
Better when participants are known and trusted. ▪ Access to n/w is controlled ▪ Governance is easy ▪ Faster transactions ▪ Better privacy and security ▪ Rules can be maintained **off-ledger** **Used when:** ▪ Banks and financial institutions are involved ▪ Enterprise or government systems ▪ Regulatory compliance is required **Example:** ▪ Bank-to-bank payment systems ▪ private blockchains	**Better when openness and decentralization are required.** ▪ Anyone can join the network ▪ No central authority ▪ High transparency ▪ Rules are fully on-ledger **Used when**: ▪ Participants are unknown ▪ Public access is needed ▪ Trust should not depend on any single entity **Example**: ▪ Bitcoin ▪ public Ethereum network

Here authority for validating transactions and issuing assets is concentrated in a single entity.

Entity A serves as the system administrator and operator of all of the arrangement's validator and asset issuer nodes.

Entity B and entity C operate proposer nodes, which gives them the ability to suggest new changes to the state of the ledger but not to make such changes without entity A's approval.

It is important to note that this is one of many different possible mappings of nodes and entities that the industry could consider.



Restricted DLT arrangement with differentiated technical roles

DLT: Potential configurations and trade-offs

Potential configurations of DLT arrangements

Table 1

Description of arrangement	One entity maintains and updates the ledger (for example, a typical FMI)	Only approved entities can use the service; entities can be assigned distinct restricted roles	Only approved entities can use the service; entities can play any role	Any entity can use the service and play any role
Operation of the arrangement	Single entity	Multiple entities		
Access to the arrangement	Restricted			Unrestricted
Technical roles of nodes	Differentiated		Not differentiated	
Validation and consensus	Within a single entity	Within a single entity or across multiple entities	Across multiple entities	

Potential Configuration and Trade-off

- **Potential configuration : Different** ways a DLT system can be set up (centralized, permissioned, or permissionless).
- **Trade-off**: A balance where improving one feature (speed, control, privacy) reduces another (decentralization, openness, trust).

In one line:

It means choosing a DLT setup by balancing **control, performance, security, and decentralization** based on needs.